

MD. Mehedi Hasan Moon

Graduate Student, Shahjalal University of Science and Technology (SUST)-3114

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EDUCATION

Master of Science (MS) in Physics (Mixed Mode)

06/2025-Ongoing

Shahjalal University of Science and Technology, Sylhet, Bangladesh

- **Thesis Title:** Computational Design and Characterization of π -Stacked Donor-Acceptor Triads for TADF OLED Applications
- **Supervisor:** Prof. Dr. Nazia Chawdhury, Department of Physics, SUST.

Bachelor of Science (Honours) in Physics

01/2019 - 04/2025

Shahjalal University of Science and Technology, Sylhet, Bangladesh

- **CGPA:** 3.04 / 4.00
- **Project title:** Numerical solution of a boundary value problem of Legendre differential equation for P_4 using finite difference method.
- **Supervisor:** Prof. Dr. Mohammed Sujaul Haque Chowdhury, Department of Physics, SUST.

RESEARCH EXPERIENCE

Master's Thesis

2025 – Ongoing

- **Thesis Title:** Computational Design and Characterization of π -Stacked Donor-Acceptor Triads for TADF OLED Applications
- **Supervisor:** Prof. Dr. Nazia Chawdhury, Department of Physics, SUST.

As a member of **Organic Optoelectronics Lab**, Department of Physics, SUST

- Investigating intramolecular π -stacked donor-acceptor triads with rigid "U-shaped" geometries to facilitate through-space charge transfer (TSCT).

- Researching Thermally Activated Delayed Fluorescence (TADF) to address "efficiency roll-off" in OLED devices.
- Utilizing **Gaussian 16** with dispersion-corrected functionals for ground-state configuration and geometry optimization.
- Implementing **ORCA 6.0** for high-level **DLPNO-CCSD(T)** calculations to ensure chemical accuracy in binding energy and excitation predictions.
- Performing hole-electron distribution analysis using **Multiwfn** to characterize electronic transitions.
- Optimizing molecular architectures to maintain a singlet-triplet energy gap (ΔE_{ST}) below **0.1 eV** for efficient reverse intersystem crossing (RISC).

Undergraduate Project

06/2023 -12/2023

- **Supervisor:** Prof. Dr. Mohammed Sujaul Haque Chowdhury, Department of Physics, SUST.
- **Project title:** Numerical solution of a boundary value problem of Legendre differential equation for P_4 using finite difference method.

Additional Research Project

2025- Ongoing

- **Supervisor:** Anock Somadder, Associate Professor, Department of Physics, SUST.
- **Title:** Time-Dependent Neutrino Emission from Neutron-Capture Processes: A Framework for Coupling SkyNet with MESA.

RESEARCH INTEREST

- **Primary:** Organic Optoelectronics, Through-Space Charge Transfer (TSCT), and TADF Emitters.
- **Secondary:** Organic Photovoltaics (OPVs), Charge Carrier Mobility in π -Stacked Systems, Exciton Dissociation and Transport, Donor-Acceptor Bulk Heterojunctions (BHJ).
- **Technical:** Computational Materials Science, Density Functional Theory (DFT), and Molecular Structure-Property Relationships.
- **Broad Interests:** Organic Semiconductors, π -Stacked Functional Materials, Solar Energy, Energy storage, Heat transfer and Nonlinear Optics.

RESEARCH WORKSHOPS

Participant | January 07, 2026

Research Workshop: The Path to Academic Excellence, SUST Research Society (SUSTRS), SUST

- Trained in research ethics, integrity, and strategies for publishing in high-impact (Q1/Q2) journals.
- Gained technical expertise in structuring research papers and using **Zotero** for citation management.

TECHNICAL SKILLS

- **Computational Software:** ORCA 6.0 | Gaussian 16 | Multiwfn
- **Programming:** Python | C++ | Bash and shell scripting | Mathematica
- **Library:** Matplotlib
- **Others:** Linux System | Word | Excel | PowerPoint | Git & GitHub
- **Theoretical Frameworks:** Density Functional Theory (DFT) | TD-DFT | Polarizable Continuum Model (PCM) simulations
- **Languages:** Bengali | English

HONORS AND AWARDS

- **Government Scholarship:**

Junior School Certificate – 2012

Secondary School Certificate – 2015

TEACHING EXPERIENCE

- **Supplementary Instructor:**
 - Facilitated comprehensive Physics and Mathematics instruction for both secondary and post-secondary students.
 - Designed and executed dynamic lesson plans featuring experiential learning and practical applications to bridge the gap between theory and reality.

- Evaluated student coursework and provided constructive, actionable feedback to drive academic growth and measurable performance improvements.

INSTITUTIONAL SERVICE

Convener | 18 December 2025

International Conference for Advances in Physics, Department of Physics, SUST

- Led the organizing committee in managing logistics, scheduling, and technical sessions for international participants.
- Collaborated with department faculty to ensure the successful execution of the conference's academic objectives.

REFERENCES

- **Prof. Dr. Nazia Chawdhury**

Professor, Department of Physics

Shahjalal University of Science and Technology,

Sylhet-3114, Bangladesh

Email: nc-phy@sust.edu

- **Prof. Dr. Mohammed Sujaul Haque Chowdhury**

Professor, Department of Physics

Shahjalal University of Science and Technology,

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Email: s.chowdhury-phy@sust.edu

- **Anock Somadder**

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